

Correction section 2.9

May 3, 2026

Exercise 2.30

Let's recall the two theorems in order to compare them:

THEOREM 2.1.33 (Change of variables). *Let J be an interval, $\varphi \in J^I$ a bijection of class $\mathcal{C}^1$ and $f \in \mathbb{R}^J$. The following assertions are true:*

$$\diamond f \in \mathcal{L}^0(J; \mathbb{R}_+) \Leftrightarrow (f \circ \varphi) \times |\varphi'| \in \mathcal{L}^0(I; \mathbb{R}_+),$$

$$\diamond f \in \mathcal{L}^1(J; \mathbb{R}) \Leftrightarrow (f \circ \varphi) \times |\varphi'| \in \mathcal{L}^1(I; \mathbb{R}).$$

Moreover if $f \in \mathcal{L}^0(J; \mathbb{R}_+) \cup \mathcal{L}^1(J; \mathbb{R})$, then

$$\int_J f(y) dy = \int_I f(\varphi(x)) |\varphi'(x)| dx. \quad (2.1.6)$$

Figure 1: First semester theorem

THEOREM 4.2.25 (Change of variables in $\mathbb{R}^d$). *Let D_1, D_2 be open subsets of $\mathbb{R}^d$, $\Phi : D_1 \rightarrow D_2$ a $\mathcal{C}^1$ -diffeomorphism and $f : D_2 \rightarrow \mathbb{R}$ a function. The following assertions are true:*

$$\diamond f \text{ is positive and measurable on } D_2 \Leftrightarrow (f \circ \Phi) |\det J_\Phi| \text{ is positive and measurable on } D_1,$$

$$\diamond f \text{ is integrable}^6 \text{ on } D_2 \Leftrightarrow (f \circ \Phi) |\det J_\Phi| \text{ is integrable on } D_1$$

where J_Φ is the Jacobian matrix of Φ . Moreover, if $f \in \mathcal{L}^0(D_2; \mathbb{R}_+) \cup \mathcal{L}^1(D_2, \mathcal{B}(E_2), \lambda^{\otimes d})$, we have

$$\int_{D_2} f(y) dy = \int_{D_1} f(\Phi(x)) |\det J_\Phi(x)| dx.$$

Figure 2: Second semester theorem

Comparison:

- " $|\varphi'|$ " becomes " $|\det(J_\Phi)|$ ". Observe that in the theorem 4.2.25, if $d = 1$, then $|\det(J_\Phi)| = |\det((\Phi'))| = |\Phi'|$: it is coherent.

- " φ bijection of class $\mathcal{C}^1$ " becomes " Φ $\mathcal{C}^1$ -diffeomorphism". It comes from the fact that there are theorems in the one dimensionnal case which ensure that if φ is $\mathcal{C}^1$, bijective and $\forall x \in I, \varphi'(x) \neq 0$, then φ^{-1} is of class $\mathcal{C}^1$ too. This does not exist in higher dimension, and that's why one has to impose that Φ^{-1} is $\mathcal{C}^1$ when $d > 1$.

Exercise 2.31

1a. Define $f_1 : \begin{matrix} \mathbb{R}_*^+ \times]-\pi, \pi[& \rightarrow & \mathbb{R} \\ (r, \theta) & \mapsto & r \cos(\theta) \end{matrix}$ and $f_2 : \begin{matrix} \mathbb{R}_*^+ \times]-\pi, \pi[& \rightarrow & \mathbb{R} \\ (r, \theta) & \mapsto & r \sin(\theta) \end{matrix}$, which are both $\mathcal{C}^1$. Then $f = (f_1, f_2)$ and :

$$\forall (r, \theta) \in \mathbb{R}_*^+ \times]-\pi, \pi[, J_f(r, \theta) = \begin{pmatrix} \frac{\partial f_1}{\partial r}(r, \theta) & \frac{\partial f_1}{\partial \theta}(r, \theta) \\ \frac{\partial f_2}{\partial r}(r, \theta) & \frac{\partial f_2}{\partial \theta}(r, \theta) \end{pmatrix}$$

thus

$$\forall (r, \theta) \in \mathbb{R}_*^+ \times]-\pi, \pi[, J_f(r, \theta) = \begin{pmatrix} \cos(\theta) & -r \sin(\theta) \\ \sin(\theta) & r \cos(\theta) \end{pmatrix}$$

and $\forall (r, \theta) \in \mathbb{R}_*^+ \times]-\pi, \pi[, \det(J_f(r, \theta)) = \cos(\theta) \cdot r \cos(\theta) - \sin(\theta) \cdot (-r \sin(\theta)) = r(\cos^2(\theta) + \sin^2(\theta)) = r$.

1b. Define $\Omega = \mathbb{R}_+^* \times]0, \pi[\times]0, 2\pi[$ and: $g_1 : \begin{matrix} \Omega & \rightarrow & \mathbb{R} \\ (r, \theta, \varphi) & \mapsto & r \sin(\theta) \cos(\varphi) \end{matrix}$, $g_2 : \begin{matrix} \Omega & \rightarrow & \mathbb{R} \\ (r, \theta, \varphi) & \mapsto & r \sin(\theta) \sin(\varphi) \end{matrix}$ and $g_3 : \begin{matrix} \Omega & \rightarrow & \mathbb{R} \\ (r, \theta, \varphi) & \mapsto & r \cos(\theta) \end{matrix}$. Then $g = (g_1, g_2, g_3)$ and:

$$\forall (r, \theta, \varphi) \in \Omega, J_g(r, \theta, \varphi) = \begin{pmatrix} \frac{\partial g_1}{\partial r}(r, \theta, \varphi) & \frac{\partial g_1}{\partial \theta}(r, \theta, \varphi) & \frac{\partial g_1}{\partial \varphi}(r, \theta, \varphi) \\ \frac{\partial g_2}{\partial r}(r, \theta, \varphi) & \frac{\partial g_2}{\partial \theta}(r, \theta, \varphi) & \frac{\partial g_2}{\partial \varphi}(r, \theta, \varphi) \\ \frac{\partial g_3}{\partial r}(r, \theta, \varphi) & \frac{\partial g_3}{\partial \theta}(r, \theta, \varphi) & \frac{\partial g_3}{\partial \varphi}(r, \theta, \varphi) \end{pmatrix},$$

thus

$$\forall (r, \theta, \varphi) \in \Omega, J_g(r, \theta, \varphi) = \begin{pmatrix} \sin(\theta) \cos(\varphi) & r \cos(\theta) \cos(\varphi) & -r \sin(\theta) \sin(\varphi) \\ \sin(\theta) \sin(\varphi) & r \cos(\theta) \sin(\varphi) & r \sin(\theta) \cos(\varphi) \\ \cos(\theta) & -r \sin(\theta) & 0 \end{pmatrix}.$$

With the formula of the 3×3 determinant,

$$\forall (r, \theta, \varphi) \in \Omega, \det(J_g(r, \theta, \varphi)) = 0 + r \cos(\theta) \cos(\varphi) r \sin(\theta) \cos(\varphi) \cos(\theta) + (-r \sin(\theta) \sin(\varphi)) \sin(\theta) \sin(\varphi) (-r \sin(\theta)) - (-r \sin(\theta) \sin(\varphi)) r \cos(\theta) \sin(\varphi) \cos(\theta) - 0 - \sin(\theta) \cos(\varphi) r \sin(\theta) \cos(\varphi) (-r \sin(\theta)).$$

With algebraic manipulations, $\forall (r, \theta, \varphi) \in \Omega, \det(J_g(r, \theta, \varphi)) = r^2 \sin(\theta)((\cos^2 + \sin^2)(\theta))((\cos^2 + \sin^2)(\varphi))$, and using $\cos^2 + \sin^2 = 1$ finally gives $\forall (r, \theta, \varphi) \in \Omega, \det(J_g(r, \theta, \varphi)) = r \cdot r \sin(\theta)$.

2. These functions refer to polar and spheric coordiantes and the expressions found correspond with the corollaries 4.2.26 and 4.2.27.

Exercise 2.32

1a. $x \mapsto \frac{1}{x^\alpha}$ is measurable (it is continuous) and positive, then the integral is well-defined (Although it is not necessarily finite!)

1b. $\varphi : \begin{matrix} I & \rightarrow & J \\ x & \mapsto & \frac{1}{x} \end{matrix}$, with $I =]1, +\infty[$ and $J =]0, 1[$, is of class $\mathcal{C}^1$ and bijective, while $f_\alpha \in \mathbb{R}^J$.

Thus, with a change of variables,

$$\int_J \frac{1}{x^\alpha} dx = \int_I \frac{1}{(\frac{1}{x})^\alpha} \left| -\frac{1}{x^2} \right| dx,$$

i.e.

$$\int_0^1 f_\alpha(x) dx = \int_1^{+\infty} x^{\alpha-2} dx.$$

1b. The preceding computations provides that $\int_0^1 f_\alpha(x) dx < +\infty$ if and only if $\alpha - 2 > -1$, i.e. f_α is integrable with respect to λ if and only if $\alpha < 1$.

2a. g_α is measurable (it is continuous as the inverse of a continuous application) and positive, thus the integral is well-defined.

2b. Let's define

$$\Phi : \begin{array}{ccc} D_1 & \rightarrow &]0, 1]^2 \\ (r, \theta) & \mapsto & (r \cos(\theta), r \sin(\theta)) \end{array},$$

with $D_2 =]0, 1]^2$ and $D_1 = \Phi^{-1}(]0, 1]^2 \cap \mathbb{R}_+^* \times]0, \frac{\pi}{2}[$. (The domain D_1 is not quite adapted to polar coordinates, thus one has to be careful with the definitions of the domains. We will have to compute D_1 in the next question.) Φ is of class $\mathcal{C}^1$ and the domains have been chosen such that Φ is bijective. Thus, one can apply the change of variables theorem to obtain:

$$\int_{]0, 1]^2} g_\alpha(x) dx = \int_{D_1} \frac{1}{r^\alpha} r dr d\theta.$$

2c. For this question, we need to compute D_1 . Let's determine it. Let $(r, \theta) \in D_1$. Then $r \cos(\theta), r \sin(\theta) \in]0, 1[$. If $r < 1$, as $|\cos| \leq 1$ and $|\sin| \leq 1$, it works no matter what is θ . If $r \geq 1$: one has $(r \cos(\theta))^2 + (r \sin(\theta))^2 < 1$, thus $r \leq \sqrt{2}$. Then, $\cos(\theta) < \frac{1}{r}$ and $\sin(\theta) < \frac{1}{r}$, which together yields $\theta \in]\arccos(\frac{1}{r}), \arcsin(\frac{1}{r})[$ (Recall that $\arcsin$ is increasing and $\arccos$ is decreasing, and everything works because $\theta \in]0, \frac{\pi}{2}[$). Therefore, if one defines $E_1 :=]0, 1[\times]0, \frac{\pi}{2}[$ and $E_2 := \{(r, \theta) \in \mathbb{R}^2 | r \in [1, \sqrt{2}[\text{ and } \theta \in]\arccos(\frac{1}{r}), \arcsin(\frac{1}{r})[\}$ (which are disjoint), $(r, \theta) \in E_1 \cup E_2$. One can reciprocally show that $E_1 \cup E_2 \subset D_1$: finally, $D_1 = E_1 \cup E_2$.

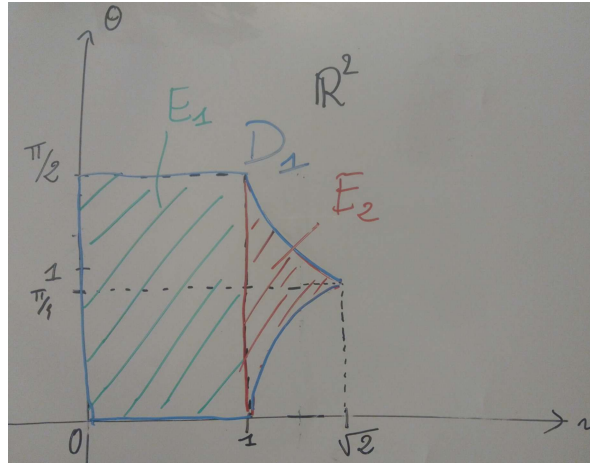


Figure 3: Representation of D_1

The fact that E_1 and E_2 are disjoint suggests the following Chasles relation:

$$\int_{]0, 1]^2} g_\alpha(x) dx = \int_{E_1} r^{1-\alpha} dr d\theta + \int_{E_2} r^{1-\alpha} dr d\theta$$

The function $(r, \theta) \mapsto r^{1-\alpha}$ is well-defined on $cl(E_2)$ and continuous, thus it is bounded by a constant $M \in \mathbb{R}_+$ (indeed, $cl(E_2)$ is bounded and closed, thus it is compact, and $\forall (r, \theta) \in E_2, r > 0$). Thus

$$\int_{E_2} r^{1-\alpha} dr d\theta \leq \int_{E_2} M dr d\theta \leq M \int_{]1, \sqrt{2}[\times]0, \frac{\pi}{2}[} \leq (\sqrt{2} - 1) \frac{\pi}{2} M < +\infty.$$

Now, let's examine the integral on $E_1 =]0, 1[\times]0, \frac{\pi}{2}[$. As the function $(r, \theta) \mapsto r^{1-\alpha}$ is measurable and positive, it is possible to use Fubini's theorem and:

$$\begin{aligned} \int_{E_1} r^{1-\alpha} dr d\theta &= \int_{]0, 1[\times]0, \frac{\pi}{2}[} r^{1-\alpha} dr d\theta \\ [\text{Fubini}] &= \int_0^1 \left(\int_0^{\frac{\pi}{2}} 1 \cdot d\theta \right) r^{1-\alpha} dr \\ &= \frac{\pi}{2} \int_0^1 r^{1-\alpha} dr \end{aligned}$$

According to the question 1.b, this integral is finite if and only if $\alpha - 1 < 1$ and the integral on E_2 is always finite. Therefore, g_α is integrable with respect to λ^2 if and only if $\alpha < 2$.

Exercise 2.33

1a. $(x, y) \mapsto \sin(x+y)$ is measurable (it is continuous), and $\int_{]-\pi, \pi[^2} |\sin(x+y)| dx dy \leq \int_{]-\pi, \pi[^2} 1 dx dy \leq (2\pi)^2 < +\infty$, thus the function is integrable.

1b. $(x, y) \mapsto \sin(x+y) \in \mathcal{L}^1(]-\pi, \pi[^2)$, then, by Fubini's theorem:

$$\begin{aligned} \int_{]-\pi, \pi[^2} \sin(x+y) dx dy &= \int_{]-\pi, \pi[} \left(\int_{]-\pi, \pi[} \sin(x+y) dx \right) dy \\ &= \int_{]-\pi, \pi[} (-\cos(\pi+y) - (-\cos(-\pi+y))) dy \\ [\cos \text{ is periodic}] &= \int_{]-\pi, \pi[} 0 dy \\ &= 0. \end{aligned}$$

Thus, $I = 0$.

1c. Let's define $\Phi : \begin{matrix}]-\pi, \pi[^2 & \rightarrow & D \\ (x, y) & \mapsto & (x+y, y) \end{matrix}$ where $D = \{(u, v) \in \mathbb{R}^2 | u \in]-\pi, \pi[\text{ and } u \in]-\pi + v, \pi + v[\}$. Φ is of class $\mathcal{C}^1$. Indeed, if $\Phi_1 : (x, y) \mapsto x+y$ and $\Phi_2 : (x, y) \mapsto y$, $\partial_1 \Phi_1, \partial_1 \Phi_2, \partial_2 \Phi_1, \partial_2 \Phi_2$ exist and are continuous. Moreover, one can observe with a small system manipulation that $\forall (x, y) \in]-\pi, \pi[^2, \forall (u, v) \in \mathbb{R}^2, \Phi(x, y) = (u, v) \Leftrightarrow (x, y) = (u-v, v) (*)$. Thus, Φ is a bijection, because it is an injection and $(*)$ provides that $D = \Phi^> (]-\pi, \pi[^2)$, proving that it is a surjection. $(*)$ also yields the expression of $\Phi^{-1} : \forall (u, v) \in D, \Phi^{-1}(u, v) = (u-v, v)$, which is also $\mathcal{C}^1$. It finally shows that Φ is a $\mathcal{C}^1$ -diffeomorphism. Let $f : (u, v) \mapsto \sin(u)$. As $f \circ \Phi : (x, y) \mapsto \sin(x+y) \in \mathcal{L}^1(]-\pi, \pi[^2)$, it is therefore possible to perform a change of variables.

1d. One has $\forall (x, y) \in]-\pi, \pi[^2, J_\Phi(x, y) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, thus $\forall (x, y) \in]-\pi, \pi[^2, |\det(J_\Phi(x, y))| = 1$.

Thus, with a change of variables:

$$\int_{]-\pi, \pi[^2} (f \circ \Phi)(x, y) |\det(J_\Phi(x, y))| dx dy = \int_D f(u, v) du dv,$$

i.e.

$$\int_{]-\pi, \pi[^2} \sin(x+y) \times 1 dx dy = \int_D \sin(u) du dv.$$

As D is not a simple product of sets, Fubini has to be performed with indicator functions (the following computation will be similar to what has been done in the exercise 2.26, section 2.8). D is a measurable

set (because Φ^{-1} is, and $D = (\Phi^{-1})^<([\pi, \pi[^2])$), thus $\mathbb{1}_D$ is measurable. Everything is well-defined and finite as proven before, thus $(u, v) \mapsto \mathbb{1}_D(u, v) \sin(u)$ is integrable on $\mathbb{R}^2$ and one can apply Fubini's theorem:

$$\begin{aligned}
\int_D \sin(u) du dv &= \int_{\mathbb{R}^2} \mathbb{1}_D(u, v) \sin(u) du dv \\
&= \int_{\mathbb{R}^2} \mathbb{1}_{]-\pi, \pi[}(v) \mathbb{1}_{]-\pi+v, \pi+v[}(u) \sin(u) du dv \\
[\text{Fubini}] &= \int_{\mathbb{R}} \mathbb{1}_{]-\pi, \pi[}(v) \left(\int_{\mathbb{R}} \mathbb{1}_{]-\pi+v, \pi+v[}(u) \sin(u) du \right) dv \\
&= \int_{-\pi}^{\pi} \left(\int_{-\pi+v}^{\pi+v} \sin(u) du \right) dv \\
&= \int_{-\pi}^{\pi} ((-\cos)(\pi+v) - (-\cos)(-\pi+v)) dv \\
&= 0 \text{ [cos is } 2\pi\text{-periodic]}.
\end{aligned}$$

It is coherent with the question 1b : $I = 0$.

2a. $(x, y) \mapsto x \sin(xy)$ is measurable (it is continuous), and $\int_{]0, 1[\times]-\pi, \pi[} |x \sin(xy)| dx dy \leq \int_{]0, 1[\times]-\pi, \pi[} 1 dx dy \leq 1 \times 2\pi < +\infty$, thus the function is integrable.

1b. $(x, y) \mapsto x \sin(xy) \in \mathcal{L}^1([\pi, \pi[^2])$, then, by Fubini's theorem:

$$\begin{aligned}
\int_{]0, 1[\times]-\pi, \pi[} x \sin(xy) dx dy &= \int_{]0, 1[} \left(\int_{]-\pi, \pi[} x \sin(xy) dy \right) dx \\
&= \int_{]0, 1[} (-\cos(\pi x) - (-\cos(-\pi x))) dx \\
[\text{cos is even}] &= \int_{]0, 1[} 0 dy \\
&= 0.
\end{aligned}$$

Thus, $J = 0$.

1c. Let's define $D =]0, 1[\times]-\pi, \pi[$: $(x, y) \mapsto (x, xy) : D \rightarrow D$. Φ is of class $\mathcal{C}^1$ (it can be justified as in 1b.) Moreover, one can observe by dividing by x that $\forall (x, y) \in D, \forall (u, v) \in \mathbb{R}^2, \Phi(x, y) = (u, v) \Leftrightarrow (x, y) = (u, \frac{v}{u}) (*)$. Thus, Φ is a bijection, because it is a injection and $(*)$ provides that $D = \Phi^>(D)$, proving that it is a surjection. $(*)$ also yields the expression of $\Phi^{-1} : \forall (u, v) \in D, \Phi^{-1}(u, v) = (u, \frac{v}{u})$, which is also $\mathcal{C}^1$. It finally shows that Φ is a $\mathcal{C}^1$ -diffeomorphism. Let $f : (u, v) \mapsto \sin(v)$. As $f \circ \Phi : (x, y) \mapsto \sin(xy) \in \mathcal{L}^1(D)$, it is therefore possible to perform a change of variables.

1d. One has $\forall (x, y) \in D, J_{\Phi}(x, y) = \begin{pmatrix} 1 & 0 \\ y & x \end{pmatrix}$, thus $\forall (x, y) \in D, |\det(J_{\Phi}(x, y))| = |x| = x$ (recall that $x \in]0, 1[$). Thus, with a change of variables:

$$\int_D (f \circ \Phi)(x, y) |\det(J_{\Phi}(x, y))| dx dy = \int_D f(u, v) du dv,$$

i.e.

$$J = \int_D \sin(xy) x dx dy = \int_D \sin(v) du dv.$$

Here things are easier than with I : D is already a product of sets, thus, as $(u, v) \mapsto \sin(v)$ is integrable on D (it is continuous on $\text{cl}(D)$, and $\text{cl}(D)$ is compact, therefore it is bounded), Fubini's theorem may

be used:

$$\begin{aligned}
 \int_D \sin(v) du dv &= \int_{]0,1[\times]-\pi,\pi[} \sin(v) du dv \\
 [\text{Fubini}] &= \int_0^1 \left(\int_{-\pi}^{\pi} \sin(v) dv \right) du \\
 &= \int_0^1 (-\cos(\pi) - (-\cos(-\pi))) du \\
 &= 0.
 \end{aligned}$$

It is coherent with the question 2b : $J = 0$.